AQA

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname _____

Forename(s) _____

Candidate signature _____

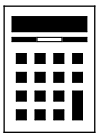
Functional Skills Certificate
FUNCTIONAL MATHEMATICS

Level 2

Monday 15 January 2018 Morning Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:
- a calculator
- mathematical instruments
- a copy of the Data Book (Examination) (enclosed).



For Examiner's Use	
Question	Mark
1	
2	
3	
4	
TOTAL	

Instructions
- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.
- State the units of your answer where appropriate.

Information
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 60.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.
- Evidence of checking is specifically assessed in Questions 2(a) and 3(a). These questions are indicated with a †.

Advice
- In all calculations, show clearly how you work out your answer.

Do not write outside the box

Answer **all** questions in the spaces provided.

1 **Dance Show**

Sue runs a dance school.



Sue

The students at my dance school are in a show.

1 (a) The show has four tap dances and one ballet dance.

Each dance has three of these students.

Amy Dita Fiona Grace Leah Mel Tia

- Amy, Dita and Tia are in the 5th dance.
- Amy is **not** in the 1st dance.
- Grace is **not** in the ballet dance.
- Each student is in at least two dances.
- No student is in consecutive dances.

Show one possible plan for the students.

[5 marks]

Do not write outside the box

Practise on this grid.

Dance	Type	Students		
1st	Tap			
2nd	Tap			
3rd	Ballet			
4th	Tap			
5th	Tap			

Put your answer on this grid.

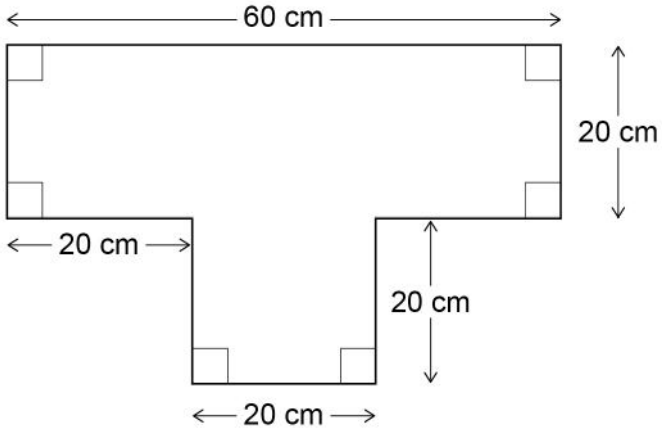
Dance	Type	Students		
1st	Tap			
2nd	Tap			
3rd	Ballet			
4th	Tap			
5th	Tap			

Question 1 continues on the next page

Do not write outside the box

1 (b) Sue makes costumes for the show.

She wants to cut out 10 of these shapes from a piece of material.



Not drawn accurately

Each scale drawing opposite shows the piece of material.

Sue says,

"I can cut out 10 of these shapes from this piece of material."

Is she correct?

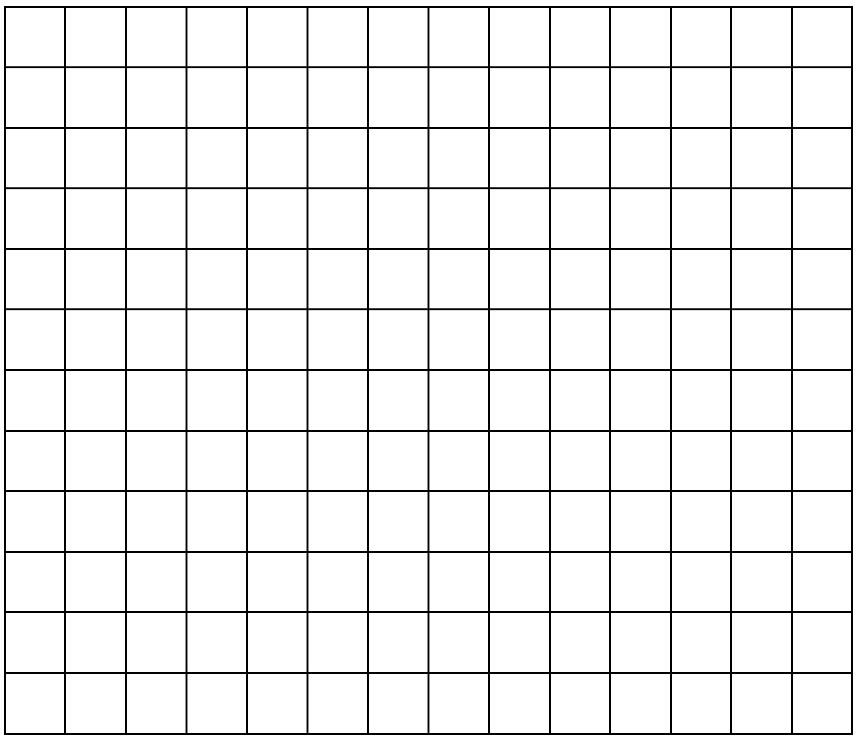
Use the scale drawings to show your working.

[5 marks]

Do not write outside the box

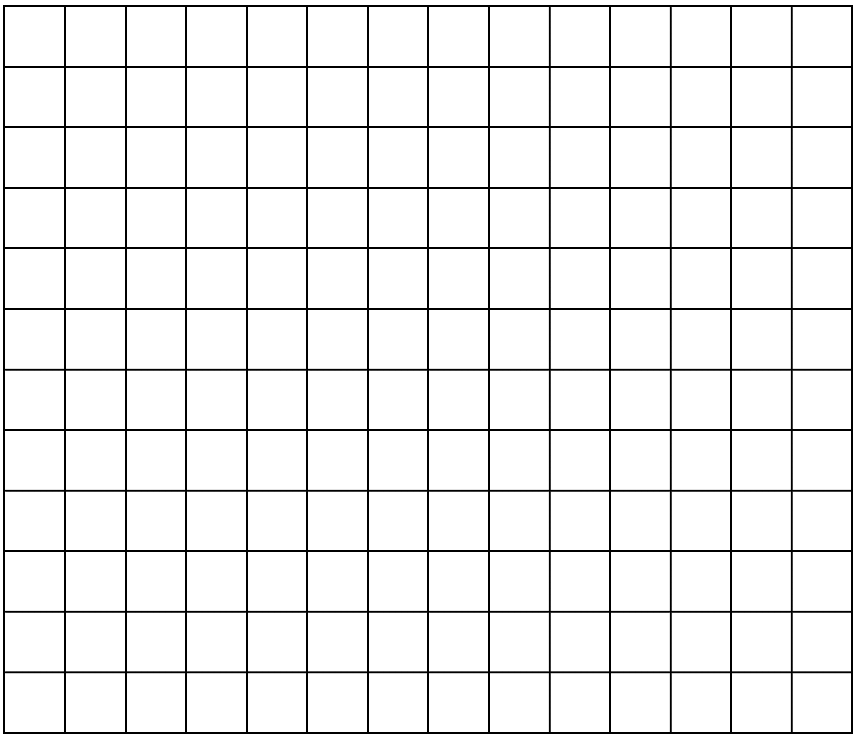
Practise on this scale drawing.

Scale ☐ represents a 10 cm by 10 cm square



Put your answer on this scale drawing.

Scale ☐ represents a 10 cm by 10 cm square



Do not write outside the box

1 (c) The show is on for **two** nights.

For each night, there are

80 adult tickets at £10.50 each

and

60 child tickets at £7.20 each.

The **total** cost of putting on the show for two nights is £925

Sue sells 90% of the adult tickets and $\frac{2}{3}$ of the child tickets.

She says,

"We have made **more than** £1200 profit."

Is she correct?

You **must** show your working.

[7 marks]

17

Do not write outside the box

2 **Minibus**

There is a **data sheet** for Minibus.

†2 (a) Samir wants to hire some minibuses to go to a hockey match.



I need enough minibuses to carry 96 people.

Samir

Each minibus can carry 15 people.

How many minibuses does Samir need to hire?

[2 marks]

Check your answer.

Show how you have done your check.

[1 mark]

Question 2 continues on the next page

Do not write outside the box

14 friends are going to Truro for a weekend.

12 of the friends will travel from Southampton.
They will pick up the other 2 friends in Bristol.

2 (b) Circle the total number of miles for the journey to Truro.

Southampton ⟶ Bristol ⟶ Truro

[1 mark]

214 268 298 536

2 (c) The friends look at this advert.

Minibus hire
£28 per day
plus
65p per mile

They drive the minibus for 3 days.

On Friday the minibus will travel

Southampton ⟶ Bristol ⟶ Truro

On Sunday the minibus will travel

Truro ⟶ Bristol ⟶ Southampton

Each friend from Southampton will pay £10 more than each friend from Bristol.

Will each friend from Southampton pay **less than** £35?
You **must** show your working.

[7 marks]

Do not write outside the box

Question 2 continues on the next page

Do not write outside the box

2 (d) On the journey, they come to a low bridge.



The height of the minibus with a loaded roof rack is 3.1 metres.

Will the minibus pass under the bridge?

You **must** show your working.

[4 marks]

15

Do not write outside the box

Turn over for the next question

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ANSWER IN THE SPACES PROVIDED

Do not write outside the box

3 **Guide Dogs in training**

Guide Dogs train at a centre each day from Monday to Friday.
In the evenings and at weekends, people look after them at home.



Sarah looks after Buddy.

Sarah takes Buddy from her home to the centre.
She then travels to work.
Sarah makes these notes.

Home to centre	10 miles average speed 30 miles per hour
At centre	5 minutes
Centre to work	$\frac{3}{4}$ of an hour

Do not write outside the box

†3 (a) How many minutes does Sarah take to travel from her home to the **centre**?

[2 marks]

Check your answer.

Show how you have done your check.

[1 mark]

3 (b) Sarah needs to arrive at **work** by 9.00 am

Work out the **latest** time that she can leave home.

[3 marks]

Question 3 continues on the next page

Do not write outside the box

The daily amount of food for a dog depends on the weight of the dog.

Weight of dog	Daily amount of food
10 kg	176 g
20 kg	296 g
30 kg	402 g
40 kg	494 g

25 kg is halfway between 20 kg and 30 kg
The daily amount of food for a 25 kg dog is 349 g

Dogs get two feeds each day.

Morning feed Half the daily amount

Evening feed Half the daily amount

3 (c) Buddy weighs 35 kg

Sarah uses a cup to measure out the **morning** feed.

1 cup holds 96 g

Sarah says,

"I need to measure out between $2\frac{1}{4}$ and $2\frac{1}{2}$ cups."

Is she correct?

You **must** show your working.

[5 marks]

Do not write outside the box

3 (d) Omar looks after a different dog.

The **daily** amount of food for his dog is 349 g

Omar says,

"A 20 kg bag of food is enough to feed my dog for 60 days.''

Is he correct?

You **must** show your working.

[4 marks]

15

Turn over for the next question

Do not write outside the box

4 **Wages**

There is a **data sheet** for Wages.



My company makes T-shirts.

Bob

The table shows the number of T-shirts 20 piece workers made in one hour.

Number of T-shirts	Number of workers
7	9
8	4
9	5
10	2
Total	20

Do not write outside the box

4 (a) Show that the mean number of T-shirts made per worker is 8

[3 marks]

4 (b) The workers are all 25 years old or over.

Bob says,

''The fair rate is £1.05 per T-shirt.''

Is he correct?

You **must** show your working.

[5 marks]

Do not write outside the box

4 (c) One week, the company needs to make at least 3600 T-shirts.

The mean number of T-shirts each worker makes per hour is 8

Bob will use

10 full-time workers, each working 7 hours per day for 5 days

and

some part-time workers, each working for $4\frac{1}{2}$ hours per day for 4 days.

How many part-time workers does he need?

You **must** show your working.

[5 marks]

13

END OF QUESTIONS

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